

KLINCAD

Definitive Manual of Electronics and PCB Design

LEARN TO DESIGN YOUR OWN CIRCUITS FROM SCRATCH

What will you find here?

A practical and super-straightforward guide to understanding how electricity works, mastering all of KlinCAD's features, using keyboard shortcuts like a professional, and fabricating your own printed circuit boards.

Software:	KlinCAD (Open Source)
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Target Audience:	High school and technical school students

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1. What is Electricity and How Does It Travel?

Before connecting components on the computer, we need to understand what happens inside the wires. It's very easy to imagine!

1.1 Atoms, Electrons, and Current

Everything around us is made up of small pieces called **atoms**. At the center of each atom is a nucleus, and spinning around it are tiny negatively charged particles called **electrons**.

In metals (like copper), electrons can easily jump from one atom to another.

- **Electric current:** It is an **ordered flow of electrons** traveling through a conductive material.
- **Circuit:** It is the closed path through which those electrons travel. If the path is cut, the current stops flowing.

The Ideal Circuit

Imagine a circular bumper car track. The electrons are the cars going around in circles. If you remove a piece of track, the cars collide and stop. For electricity to perform a useful task, it needs a continuous round-trip path without cuts.

1.2 What is a PCB Board?

When we build a circuit on a breadboard, we use many loose wires that can accidentally disconnect. A **PCB** (*Printed Circuit Board*) is a rigid board where the wires are drawn as fixed copper traces. KlintCAD is the program that helps you design these boards.

2. The Laws that Rule Electronics

For your circuits to work well, there is a key rule you must remember: **Ohm's Law**.

2.1 Voltage, Current, and Resistance

Imagine electricity like water in a hose:

- **Voltage (V):** It is the pressure that pushes the electrons (in Volts, V).

- **Current (I):** It is the amount of electrons passing per second (in Amperes, A).
- **Resistance (R):** It is the opposition to the flow of current (in Ohms, Ω).

The basic formula is:

$$V = I \cdot R \quad (2.1)$$

If you raise the voltage too much and leave little resistance, the current will be huge and the component will burn out. That is why we always use resistors to protect delicate parts like LEDs.

3. Getting to Know KlinCAD: Menus, Options, and the 4 Layers

KlinCAD organizes printed circuit board design through an intuitive interface with 4 working layers and agile configuration tools.

3.1 The 4 Design Layers

To avoid unwanted crossings between components, the program separates the work into color-differentiated layers:

1. **Top Copper (Upper layer - Red):** Routing of copper traces on the top face of the board.
2. **Bottom Copper (Lower layer - Blue):** Traces on the bottom face (most used in single-sided boards).
3. **General / Jumpers (Green):** Guides, notes, and wire jumpers when two traces must cross.
4. **Power / Silkscreen (Orange):** Marking of ground lines (*GND*), voltage, and informational texts.

3.2 Menus, Settings, and Export Formats

The top menu allows you to customize the working environment and export your final blueprints:

- **Language Selector (ES / EN) and Dark Mode:** Real-time bilingual support (LANG_DICT) and dark interface option to reduce visual fatigue.
- **Native file (.klincad):** Main editable project format.
- **Export to PDF (Ironing Mode):** Allows printing the design at actual size or mirrored for home thermal transfer.
- **Export to PNG:** Clean, high-resolution image ideal for submitting technical reports.

3.3 Integrated Calculator and Component Connection

- **Electronic Calculator:** Menu for quick value retrieval according to Ohm's Law ($V = I \cdot R$), Power ($P = V \cdot I$), and Resistivity Coefficient without resorting to external software.
- **Circuit Connection:** Place components on the grid and route traces between their pins. To ensure proper operation, verify that the current travels in a direct closed path from the source to ground (*GND*).

4. Keyboard Shortcuts and Agile Navigation

To work quickly and comfortably, KlinCAD features essential keyboard shortcuts that you should memorize:

4.1 Main Shortcuts and Quick Editing

- **Ctrl + S:** Saves the current project.
- **Ctrl + Z and Ctrl + Y:** Undo and Redo actions.
- **Esc:** Instantly returns to pointer mode to select or move elements.
- **X:** Activates the tool to draw copper traces.
- **Shift + X:** Toggles special trace routing modes.
- **R:** Rotates selected components on the grid.
- **Delete / Backspace:** Deletes selected components or traces.
- **Keys 1, 2, 3, and 4:** Select the active layer (1 Top, 2 Bottom, 3 Jumpers, 4 Power).
- **Ctrl + C / Ctrl + V / Ctrl + X:** Copy, paste, and cut elements.

4.2 Zoom, Panning, and 2.5D Viewer

- **Zoom with the wheel:** Hold down **Ctrl** and scroll the mouse wheel to zoom in or out.
- **Panning:** Hold down the right mouse **click** and drag to move the canvas. It works identically in the **2.5D interactive viewer**.
- **Instant Information (Tooltips):** When hovering the cursor over any component, KlinCAD displays a brief functional description.

Don't forget to save!

Get into the habit of pressing **Ctrl + S** constantly to avoid losing your progress.

5. Error Verification, Bill of Materials, and Catalog

5.1 Error Checking (DRC) and Bill of Materials (BOM)

To guarantee a functional design before manufacturing, KlinCAD integrates auditing tools into its code:

- **DRC (Design Rule Check / Project Errors):** It is the program's technical verification algorithm. It examines the logical pins (`pinos_logicos`) of each component and checks their electrical continuity with traces (`PistaInteractiva`) and nodes (`Nodo`). If it detects disconnected pins or incomplete circuits, it immediately warns the user before export.
- **Visual Error Arrows (`FlechaError`):** When running the DRC, the system generates red arrow-shaped graphic polygons over the coordinates of the faulty pin. They indicate the existence of a "Floating Pin" or connection error. They are interactive elements that the student can inspect, correct by routing the missing trace, or delete manually (`Supr`).
- **BOM (Bill of Materials):** Scans the circuit and counts all instances of the `ComponenteLiviano` class. It groups elements by their identifier (`ref_id` like R_1 , C_1 , U_1), their type, and their commercial value (`valor`). It allows students to know the exact list of supplies to check stock at the school storage or make the purchase prior to assembly.

5.2 Components You Will Use

In the KlinCAD library you will find the fundamental parts and their representative symbols to build your practices:

- **Resistors:** Limits the flow of electrical current protecting other components. (*Symbol:* Rectangle with terminals).
- **Capacitors:** Store electrical energy temporarily and filter noise in lines. (*Symbol:* Two separate parallel plates).
- **Inductors / Coils:** Store energy in a magnetic field through a wire winding. (*Symbol:* Series spirals).
- **Diodes:** Allow current flow in only one direction. (*Symbol:* Triangle pointing to a vertical line).
- **Zener:** Special diode that maintains a constant voltage in reverse bias for regulation. (*Symbol:* Triangle with Z-bar).
- **Transistors:** Function as fast electronic switches or amplifiers. (*Symbol:* Circle with Base, Collector, and Emitter).
- **Integrated Circuits (ICs):** Chips that group complex logical or analog functions. (*Symbol:* Block with side pins).
- **LEDs:** Light-emitting diodes. (*Symbol:* Diode with light radiation arrows).

- **Buzzers:** Electroacoustic devices that generate sound or audible alerts. (*Symbol:* Two-terminal transducer).
- **7-Segment Displays:** LED arrays to visualize numbers. (*Symbol:* Box with segments labeled from A to G).
- **Push Buttons:** Buttons that close a circuit momentarily. (*Symbol:* Mechanical contact with tab).
- **Switches:** Devices to open or close circuits permanently. (*Symbol:* Toggle lever).
- **Batteries / Cells:** Portable sources that provide electrical energy to the circuit. (*Symbol:* Parallel lines of alternating polarity).
- **Power Terminals:** Fixed connection points for specific voltages or ground (*GND*). (*Symbol:* Voltage arrow or ground symbol).
- **Board Connectors:** Physical interfaces (screw terminals or pins) for external cables. (*Symbol:* Block with input terminals).
- **Mounting Holes:** Mechanical perforations to secure the PCB to the enclosure. (*Symbol:* Circle with fixation area).

6. Help, Community, and License

6.1 Help Menu and Discord Community

If at any time you have doubts about how to use a function or want to propose improvements, the program features a **Help Section**. There you will find direct links to:

- Consult additional guides and documentation.
- Join the **Discord community** to chat with other students and report bugs.

6.2 Open Source and License

KlinCAD is an **open source** development. This means you can use, study, and share it freely.

License and Authorship Notice

If you are going to use this code, modify it, or base your project on it, by license rules you must always include recognition of its original author: **Matias Baranovsky**.